# Live Sound and Recording Techniques
# Assignment 2: Live recording project

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## Preparation (format and equipment)

We started e-mailing bands early to try and have time to prepare, getting a positive reply from Pete Molinari who was playing in Bush Hall. We thought about the best way to record the gig, one possible way being to run a splitter and use my Protools setup (18 channel) but instead we decided to rent an Alesis HD24 multitrack recorder from Fx rentals as it would probably be more stable. We also called into the venue beforehand to get clearance to record.

## Mics and the band

Pete Molinari had a seven-piece rockabilly band consisting of double bass, drums, lead guitar, rhythm guitar, pedal/lap steel, lead vocals and a pair of backing singers.



The kick was mic'd with a Sure Beta 52, toms with a Shure beta 98 D/S, hats with an AKG SE300B, top snare with a Sure SM57 and bottom snare with a Sennheiser e604.

The double bass was DI'd and amped, the amp having a Sure SM57 on it. The guitars and the pedal steel also both had Sure SM57's on them. The vocals were all Sure SM58's.

We also put up our own ORTF pair of Shuniyin Cy90s at the back of the hall with the cables run above the door, behind the desk and into a pre amp that we borrowed from the MRC. Although I had a Focusrite Saffire we could have used, we borrowed a different one from the MRC because it had a soft-limiting option which would come in handy for any unexpected level jumps, this made sense as we knew the head room on the saffire was low.

## Bush Hall

We were able to obtain a lot of information about the venue  from their website. They have a 63-amp 3-phase power supply and also a 63-amp single-phase power supply off which they run the PA; but on a 32-amp tap.

They had their own multi-core laid under the floor with the front of house consisting of:

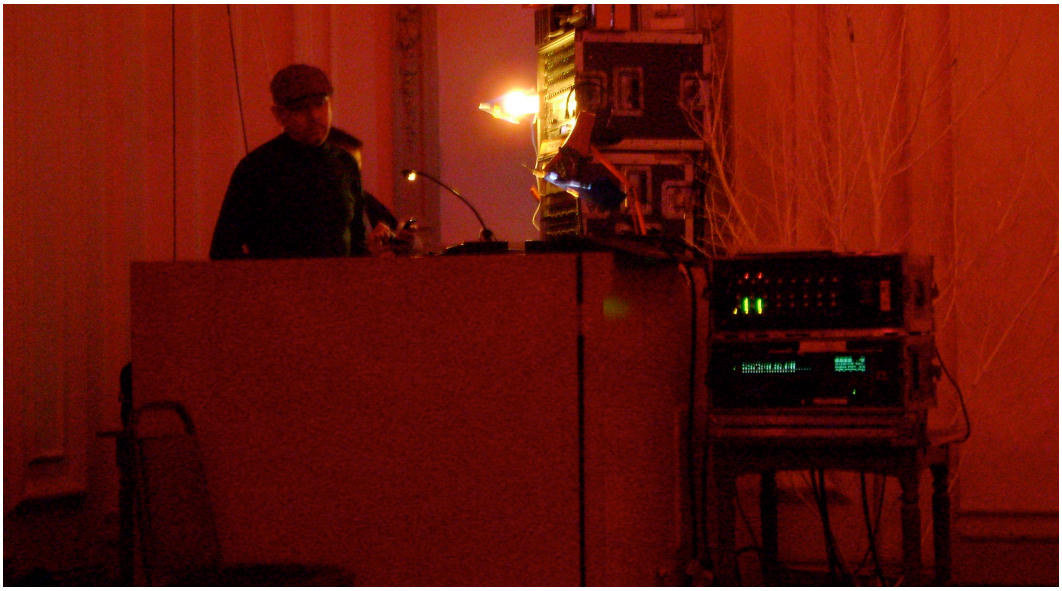
- 8  Martin Audio W8LN Line Array Speakers,
- 4  Martin Audio WMX Sub Bass Speakers,
- 6 Martin Audio LE1200 Monitors for fold-back monitors.

This set up was a nice PA but incorrectly set up. It is best to 'fly' a PA in order to obtain maximum angle of incidence into the thermal layer created by humans (Rat 2009). The Bass bins were stacked above the ground and although Martin Audio say its for either being ground stacked or flown, it would transmit bass better through the ground. (Martin Audio 2009)



Also analysing at the room dimensions provided they show that the typical build up of modes happen from about 51hz upwards meaning that below this would not be the most stable (and proved messy live) though because this happened at a reasonably low frequency, it was not a big problem in the mix. With the hall sufficiently full the it didn't tend to have many problems bar the low end, in this case it should also be noted that the engineers rang the room out earlier in the day.

The venue was running a Midas Venice 320 console of which we took the direct outs straight into the multitrack recorder. Unfortunately we had to have our signal coming in post fade, which meant our recording had effects such as compression, gating and level rides on every track the engineer used them on.



Luckily the engineer was reasonably good so the effect of gating the toms and bottom snare caused few problems in the mix, though Pete's rhythm guitar was compressed beyond recognition and was practically useless. The level rides were generally pretty smooth; though there were still a good few of them that had to be fixed in the mix with automation.

Pete's group had 16 of the 24 channels used at sound check while Nat Johnson and the figure heads (the warm up band who we also had permission to record) took up an additional 6 channels, paired with the ORTF ambience set up that we placed ourselves we had all 24 of our channels used. If we had run with the initial idea of using Protools idea we would not have been able to record both bands properly due to having only 18 channels rather than the 24 and would have had to come up with a compromise.

As we were guests of the venue we were not allowed to mic up the instruments our way but for the most part the signals were pretty good. In my opinion the kick drum and double bass were the worst of the signals apart from Pete's guitar, which was destroyed by the compressor. This wasn't as big an issue as it sounds considering Pete could be pushed into the background without a noticeable effect (his playing lacked), not to mention the Fender Pro Junior amp he had sounded awful. The Double bass was amped, which I found to be pretty useless in the mix but also DI'd which gave a much better sound once some heavy compression and Eq was applied. The kick sound was very boxy but with Eq and compression it sounded a lot better.

The pedal steel sound was excellent as was the lead guitar. Though the best sound was probably our ambience mic's, which allowed me to only use one reverb for the vocals at mix stage, simply pushing up the ambience for everything else. This was an ORTF pair of Shuniyin Cy90s set up at the very back of the hall, which we guarded throughout the gig while punters constantly tried to walk behind it and between the 2ft of cabling from the stand to the wall, the rest of

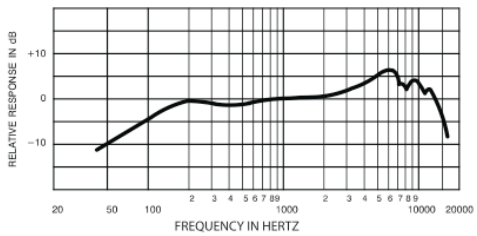
which we had ran above the door and over to the desk. This mic setup gave a very nice stereo image of both the crowd and the on stage sound; helping naturally position the musicians. **An edited example of important parts of the gig with just this ambience mic set up can be heard on track 4.*



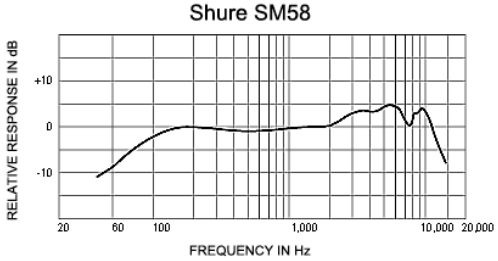

The backing vocals proved very problematic as some of the vocals were out of tune and sung early. If I had to guess; I would say the singers couldn't hear themselves in their monitor properly, also directly behind them were two guitar amps and a drummer who was slightly off to one side, meaning that there was a lot of spill on these tracks. This meant that I had to gate them with a very fine threshold during the mix. I also added reverb to try disguise the notes that were off key.

It would have been nice if super cardioids could have been used on the backing vocals to have reduced off axis spill. I would have also recommended moving the singers somewhere quieter on stage but this was not possible considering how full it actually was.

Another consideration during mixing was that there were a lot of the same mic's on stage. These were predominantly Sm57s and Sm58s and looking at the frequency response curves of these mic's you'll notice both have a significant presence boost from around 2k. With so many of these mic's being used these frequencies tend to build up. Depending on the instrument, I tried to reduce this presence boost when I could if it wasn't a critical frequency band.




*Sure Sm57*




*Sure Sm58*

Although we recorded both bands, I chose my songs entirely from the Pete Molinari set as the band was more complex and also had better musicians. While Nat Johnson and the Figureheads were fine as a band they lacked the professionalism of the main act. I found the drummer's dynamics to be less controlled so it would have been closer to a restoration attempt rather than a mix.

## Conclusion

Throughout the process we have been prepared and planned for any possible problems, executed the task of getting a good live recording to hard disk and mixed it to a professional standard with a runtime of over 15 minutes. Although many things could have gone wrong we were prepared for most circumstances and successfully navigated problems to the benefit of our recording.

## References

Rat, D. (2009) The Thermodynamics of a Rock Show *[Online]* http://en.audiofanzine.com/articles/a.print,The-Thermodynamics-of-a-Rock-Show.html Viewed 09/01/10

Martin Audio (2009) Martin audio ltd. *[Online]* http://www.martin-audio.com/products/WMX.asp Viewed 09/01/10

## Discography

(research listening)

The Band – The Last Waltz
John Butler Trio – Live at St. Gallen
Neil Young – Live at Fillmore East
Manfred Manns Earth Band – Wired Live (bad recordings)

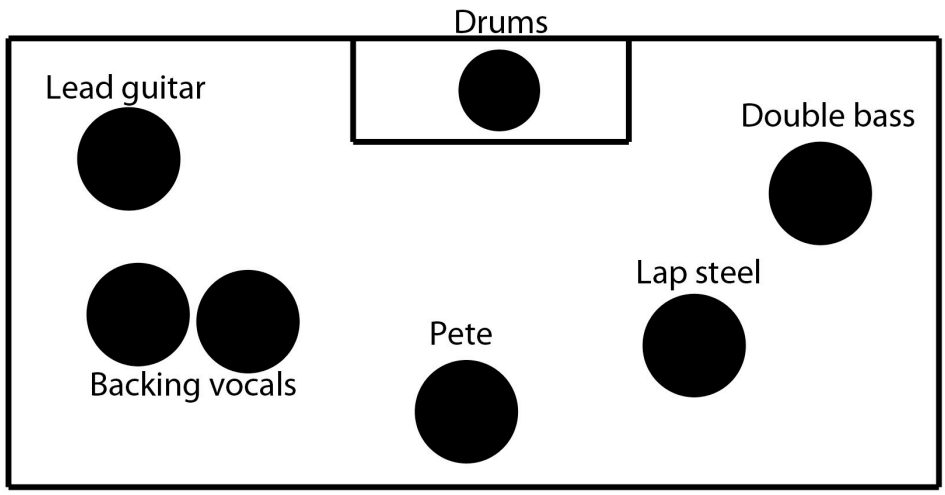
# Appendix

## *Appendix 1 (track sheet)*

| Channel | Source | Mic | Notes |
|---|---|---|---|
| 1 | Kick | Sure Beta 52 | Pete and Nat Johnson |
| 2 | Snare top | Sure Sm57 | Pete and Nat Johnson |
| 3 | Snare bottom | Sennheiser e604 | Pete and Nat Johnson |
| 4 | High hats | AKG SE300B | Pete and Nat Johnson |
| 5 | Hi tom | Sure Beta 98D/S | Pete and Nat Johnson |
| 6 | Floor tom | Sure Beta 98D/S | Pete and Nat Johnson |
| 7 | Double bass DI | DIBSS AR133 Active DI Box | Pete |
| 8 | Double bass amp | Sure Sm57 | Pete |
| 9 | Pedal steel | Sure Sm57 | Pete |
| 10 | Lead guitar | Sure Sm57 | Pete |
| 11 | Pete's guitar | Sure Sm57 | Pete |
| 12 | Backing vocals | Sure Sm58 | Pete |
| 13 | Backing vocals 2 | Sure Sm58 | Pete |
| 14 | Main vocals | Sure Sm58 | Pete |
| 15 | Resonator | Sure Sm57 | Pete |
| 16 | Acoustic guitar | Emo Passive DI | Pete |
| 17 | Ambience left | Shuniyin Cy90 | Pete and Nat Johnson |
| 18 | Ambience right | Shuniyin Cy90 | Pete and Nat Johnson |
| 19 | Bass DI | BSS AR133 Active DI Box | Nat Johnson |

| | | | |
|---|---|---|---|
| 20 | Guitar one | Sure Sm57 | Nat Johnson |
| 21 | Guitar two | Sure Sm57 | Nat Johnson |
| 22 | Nats vocals | Sure Sm58 | Nat Johnson |
| 23 | Backing vocals | Sure Sm58 | Nat Johnson |
| 24 | Glockenspiel | Sure Sm58 | Nat Johnson |

## *Appendix 2 (Stage Plan)*



# Bush Hall Signal Flow

